Macroeconomic Forecasting of U.S. Wage Growth

ECON 3916: Machine Learning Final Project

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1. Problem Statement

Prediction question. Can observable macroeconomic indicators predict the year-over-year growth rate of U.S. average weekly earnings for private-sector employees?

This is a prediction problem, not a causal one. The goal is to forecast the direction and magnitude of wage growth using currently available economic data, not to establish whether any indicator causes wages to change. The distinction matters because feature importance in a predictive model reflects correlation and predictive signal, not causal effect. Throughout this report, all references to feature importance carry that caveat explicitly.

Stakeholder and decision. This analysis would help monetary policy analysts at the Federal Reserve or labor economists at the Bureau of Labor Statistics decide whether current macro conditions signal upcoming wage acceleration or deceleration. This information is relevant to interest rate decisions, inflation forecasting, and labor market monitoring. A reliable wage growth forecast with honest uncertainty bounds would allow policymakers to act proactively rather than reactively.

Dataset. All data is sourced from the Federal Reserve Economic Data (FRED) API, accessed April 15, 2026. The full pipeline pulls 229 economic time series spanning inflation, GDP, labor markets, interest rates, money supply, housing, consumption, trade, commodities, and fiscal policy. A custom scoring layer (56 registered blocks) produces derived columns including year-over-year rates, regime flags, continuous composite scores, and lead/lag signals. The research subset contains 998 weekly observations across 34 columns, with the target variable Avg_Weekly_Earnings_YoY derived as the 52-week percent change in BLS series CES0500000011.

2. Data Description

Source and scope. The FRED API serves as the single data source, accessed through a custom FRED_Loader submodule that fetches, resamples, and scores all 229 series. All series are resampled to weekly frequency (Friday-aligned) using last-observation-carried-forward for monthly and quarterly data, and weekly means for daily data. The dataset spans January 2000 through the most recent FRED release, though the dependent variable is only available from March 2007 onward due to the 52-week lookback required for the YoY transformation.

Target variable. Average weekly earnings YoY growth is centered around 2-4% in normal periods, with a COVID-era spike to approximately 7.5% and a minimum of roughly 0.7%. The YoY transformation removes the upward non-stationary trend and aligns the target with features already expressed in rate-of-change form (e.g., Core PCE YoY, Credit Impulse).

Quality assessment. The research subset contains zero missing values. The upstream pipeline handles gaps via forward-fill at the weekly resampling stage, and a datetime trim removes forward-filled stale observations beyond the last real FRED release date. Given the forward-fill mechanism, missingness in the raw monthly/quarterly series is best classified as MAR (Missing At Random), conditional on publication schedule, not on the values themselves.

Feature set

Twelve features were selected from the original 32 via correlation analysis and stationarity checks. Three originally selected level variables (Retail Sales, JOLTS Quits, Part-Time for Economic Reasons) were replaced with stationary scored alternatives after temporal evaluation revealed that trending level variables cause extrapolation failure when the test set lies strictly in the future. The final set spans labor market conditions, inflation, financial conditions, household behavior, real activity, and autoregressive signal.

Feature	Source	Description
Core_PCE_YoY	Scored	Core PCE inflation, 52-week percent change
JOLTS_UE_Ratio	Scored	Job openings per unemployed person
Personal_Savings_Rate	FRED	Household savings as % of disposable income
Labor_Force_Participation	FRED	Civilian labor force participation rate
Sahm_Indicator	Scored	3-month UE avg minus 12-month UE min
HY_OAS_Spread	FRED	High-yield corporate bond OAS spread
U6_Underemployment	FRED	Broad underemployment rate (U-6)
Employment_Pop_Ratio	FRED	Employed persons / civilian population
Payroll_Change_4w	Scored	4-week change in nonfarm payrolls
Activity_Momentum	Scored	Composite z-score: IP, retail, PCE, payrolls
Earnings_YoY_Lag4	Derived	Target variable lagged 4 weeks
Earnings_YoY_Lag13	Derived	Target variable lagged 13 weeks

Table 1. Final feature set (12 predictors).

3. Methodology

3.1 Train/test split

The dataset is split temporally at the 80th percentile: approximately 788 training observations (March 2007 through July 2022) and 197 test observations (July 2022 through April 2026). This ensures the model is evaluated on a genuinely future macro regime, specifically the post-pandemic wage deceleration, that it has never seen during training. All random operations use random_state=42 for reproducibility. The choice of a temporal split rather than a random split is essential: random splits allow the model to train on future data that chronologically follows test observations, leaking information about regime structure and producing misleadingly high apparent performance.

3.2 Models compared

Model 1: OLS Linear Regression (baseline). Ordinary least squares with no regularization, providing a closed-form solution and serving as the interpretability baseline. Coefficients are directly readable as marginal effects on predicted wage growth.

Model 2: Ridge Regression (alpha = 1.0). Adds an L2 penalty to shrink coefficients toward zero, reducing variance at the cost of some bias. Ridge distributes coefficient magnitude more evenly across collinear predictors. In practice, Ridge performed nearly identically to OLS, indicating multicollinearity was not a dominant problem in the 12-feature set.

Model 3: Gradient Boosting Regressor. An ensemble of 200 sequential shallow decision trees (max_depth=3), each fitting the residuals of the cumulative ensemble. Learning rate = 0.03 with subsample = 0.8 for stochastic gradient boosting. The model captures nonlinear interactions and regime-dependent feature effects that linear models miss, but at the cost of interpretability and generalization under regime shift.

3.3 Cross-validation

TimeSeriesSplit with 5 folds ensures each fold trains on an expanding window of past data and tests on the next chronological segment. This prevents temporal leakage that would inflate metrics under standard k-fold CV. Each successive fold adds approximately 160 additional training observations while testing on the next chronological block.

3.4 Evaluation metrics

RMSE (root mean squared error), MAE (mean absolute error), and R-squared are reported for each model. Negative R-squared indicates the model predicts worse than simply guessing the training-set mean for every observation. RMSE penalizes large errors more heavily, while MAE provides a more robust measure of typical prediction magnitude.

4. Results

4.1 Test set performance

Model	RMSE	MAE	R-squared
OLS	0.3977	0.3169	0.011
Ridge (alpha = 1.0)	0.3965	0.3162	0.017
Gradient Boosting	0.4525	0.3763	-0.280

Table 2. Test set metrics under temporal split (with lag features).

Ridge regression achieves the best test-set performance with an R-squared of 0.017 and RMSE of 0.3965, marginally outperforming OLS. Both linear models substantially outperform Gradient Boosting, which produces a negative R-squared of -0.280 on the temporal test set. All three models perform close to or worse than the naive mean-prediction baseline, indicating that macro features provide negligible predictive power over genuinely future wage growth.

Figure 1 illustrates this visually. The OLS and Ridge predicted series (blue and teal) track the general level of actual wage growth (black) through the test window but fail to anticipate the timing or magnitude of short-run fluctuations. The GBR predictions (dashed orange) diverge more substantially, undershooting actual growth through much of 2023-2024 and exhibiting larger swings that reflect memorized training-era patterns rather than genuine foresight.

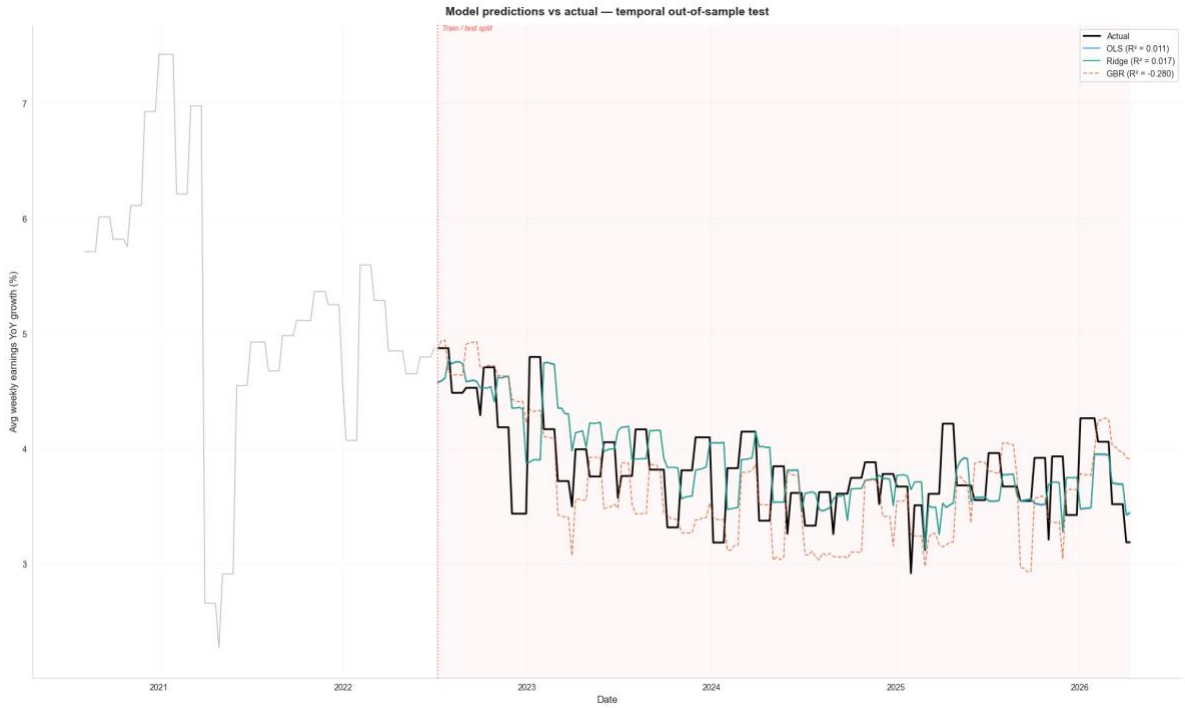

Figure 1. Actual vs. predicted wage growth, July 2022 through April 2026. Red dashed line marks the train/test split.

4.2 Cross-validation results

Model	CV R-squared (mean +/- std)
OLS	-2.92 +/- 1.92
Ridge	-2.06 +/- 1.23
Gradient Boosting	-1.51 +/- 1.21

Table 3. TimeSeriesSplit cross-validation (5 folds).

All three models produce deeply negative mean R-squared values across the five temporal folds, confirming that poor out-of-sample performance is not an artifact of the particular train/test boundary. The enormous standard deviations (ranging from +/-1.21 to +/-1.92) demonstrate that the macro-to-earnings relationship is regime-dependent: a model that performs reasonably in one temporal window can fail catastrophically in the next.

4.3 The overfitting problem: random vs. temporal evaluation

Under a standard random 80/20 train/test split (ignoring temporal order), the Gradient Boosting model achieves R-squared of 0.99, near-perfect fit. Under the temporally honest split, the same model drops to R-squared of -0.28, the worst of all three models. This 1.27-point gap demonstrates that GBR memorizes training-era patterns (such as the specific relationship between JOLTS ratios and wage growth during the 2015-2022 expansion) that invert during regime shifts. Linear models, which cannot memorize nonlinear regime-specific patterns as aggressively, extrapolate more gracefully into the post-pandemic test window. This comparison underscores a broader lesson: a model that appears to perform brilliantly under random evaluation can be entirely useless for actual forecasting.

4.4 Feature importance

Gradient Boosting Gini importance is dominated by Labor Force Participation at approximately 58% of total importance, followed by the two autoregressive lag features. U6 Underemployment is the only macro

feature contributing meaningful signal beyond autoregression. Adding the 4-week and 13-week lagged earnings features improved RMSE from approximately 0.48 (without lags) to 0.40 (with lags), confirming that the best predictor of near-term wage growth is recent wage growth itself.

Predictive importance only. Does not imply causal effect.

5. Recommendation

5.1 Actionable advice

Policymakers and labor economists should not rely on a static ML model trained on historical macro indicators as the primary tool for forecasting wage growth. The evidence from this analysis is clear: while macroeconomic features show strong contemporaneous correlation with earnings growth, that correlation does not translate into reliable out-of-sample prediction under temporally honest evaluation. A naive forecast (predicting the training-set mean) outperforms all three models on genuinely future data.

The strongest practical recommendation is to use a simple autoregressive approach: the best predictor of near-term wage growth is recent wage growth itself. Ridge regression offers marginal improvement when lag features are included (R-squared of 0.017), but the improvement is small enough that it should be treated as noisy rather than actionable.

5.2 Uncertainty quantification

The Streamlit dashboard includes a Monte Carlo forecast engine that provides a more honest picture of forecast uncertainty. The engine fits empirical marginal distributions to each of the 12 input features from training data, infers the empirical correlation matrix, and draws correlated samples via a Gaussian copula (Cholesky decomposition followed by inverse CDF transformation to each marginal). Under 10,000 correlated draws through the fitted Ridge model, the simulation produces a mean predicted wage growth of 3.127%, a median of 2.909%, and a standard deviation of 1.123 percentage points. The 95% confidence interval spans 1.495% to 6.082%.

Figure 2. Monte Carlo forecast distribution (10,000 iterations, empirical marginals). Dashed red lines mark 95% CI bounds.

The forecast distribution exhibits a notable right skew, with a long tail extending above 6%. This asymmetry reflects the potential for high-wage-growth scenarios driven by tight labor market conditions (low U6, high JOLTS ratio) that the training data includes from the 2021-2022 period. The convergence diagnostic confirms that the cumulative mean stabilizes well before the 10,000th iteration, with a suggested N of 25,000 for tighter confidence bands.

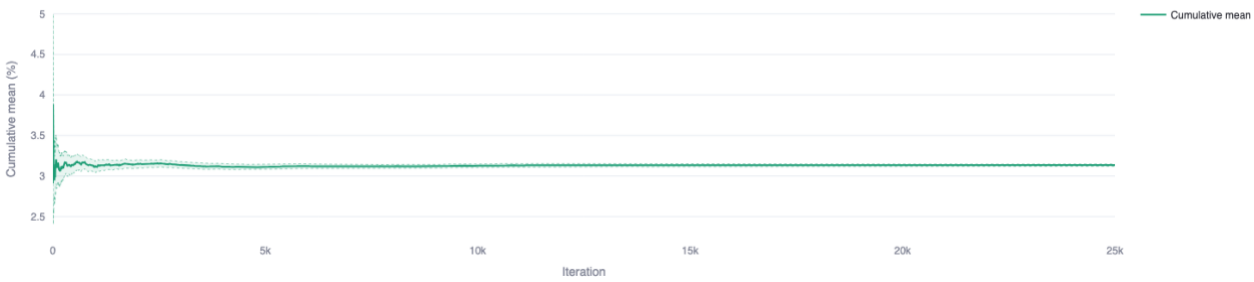

Figure 3. Convergence diagnostic. Cumulative mean stabilizes by ~5,000 iterations; suggested N = 25,000.

The wide 95% CI of approximately 4.6 percentage points (1.5% to 6.1%) reinforces the central finding of this report: macro-based wage forecasting carries substantial uncertainty, and any point estimate should be accompanied by its interval. A policymaker using this forecast would need to plan for wage growth anywhere from below 2% to above 6%, which is too broad a range to inform precise policy calibration.

5.3 Limitations

Several limitations warrant acknowledgment. First, the forward-fill resampling strategy introduces staleness in monthly and quarterly features at weekly resolution; intra-month variation is lost. Second, the 12-feature set was selected via contemporaneous correlation, not via a formal temporal feature selection procedure such as Granger causality tests. Third, the test period (mid-2022 through early 2026) represents a single macro regime: post-pandemic disinflation and labor market normalization. Performance could differ substantially in an expansion or deep-recession regime. Fourth, all models treat the macro-to-wage relationship as stationary across the training window, despite clear evidence of structural breaks such as the COVID-era spike to 7.5% YoY growth.

5.4 Next steps

Future work should explore regime-switching models that explicitly allow coefficients to vary across macro regimes, longer forecast horizons with appropriately lagged features (e.g., 13- or 26-week-ahead targets) to strengthen the predictive rather than contemporaneous signal, ensemble approaches that blend autoregressive components with macro indicators via residual stacking, and alternative target definitions such as real wage growth (deflated by CPI) or sector-specific earnings to test whether the null result generalizes beyond headline private-sector earnings.